Class Problem Set (Computer Performance)

Q1. A compiler designer is trying to decide between 2 code sequences for a particular computer. The hardware designers have supplied the following facts (*see Table 1*)

The two code sequences require different instruction count (*see Table 2*)

(i) Which code sequence executes the most instructions?

(ii) Which will be faster?

(iii) Which sequence needs lesser CPI?

Instruction Class	**CPI**
A	2
B	3
C	4

Table 1

Code Sequence	**IC for each Class**		
	A	**B**	**C**
1	1	2	3
2	4	1	2

Table 2

Q2. ISA of a machine consists of 3 different instructions types A, B and C. When benchmark program P is run on it, it is translated as per information given in the table.

Find average CPI for benchmark.

Instruction Class	**CPI**	**Frequency**
A	2	35%
B	3	25%
C	4	40%

Q3. Consider the following performance measurements for a program;

Measurement	**Computer A**	**Computer B**
Instruction Count (IC)	12 billion	9 billion
Clock Rate (CR)	6 GHz	6 GHz
Cycles/ Inst. (CPI)	1.0	1.1

a. Which computer has higher MIPS rating?

b. Which computer is faster (Execution time is lesser)?

Q4. A new computer is designed which is 5 times faster on computation in the web serving application than the original processor. Assuming that the processor is busy with computation 60% of the time and is waiting for I/O 40% of the time, what is the overall speedup gained by incorporating the enhancement?

Q5. In the embedded market, where cost is crucial, processors sometimes implement floating point only in software. We are interested in two implementations of a computer, one with and one without special floating-point hardware.

Consider a program, P, with the mix of instructions as shown in Table 1

Instructions	**Frequency**	**Cycles**	**No. of Integer Instructions**
Floating-point multiply	10%	6	30
Floating-point add	15%	4	20
Floating-point divide	05%	20	50
Integer instructions	70%	2	NA

Computer MFP (machine with floating point) has floating-point hardware and can therefore implement the floating-point operations directly. It requires the number of clock cycles for each instruction class as indicated in table 1.

Computer MNFP (machine with no floating point) has no floating-point hardware and so must emulate the floating-point operations using integer instructions. The integer instructions all take 2 clock cycles. The number of integer instructions needed to implement each of the floating-point operations is also given in the table. Both computers have a clock rate of 1GHz.

a. Calculate the MIPS ratings for both computers.

b. If the computer MFP needs 300 million instructions for this program, how many integer instructions does the computer MNFP require for the same program?

c. What is the execution time (in seconds) for the program P executed on MFP and MNFP?

d. Which machine has better performance on the basis of (i) MIPS rating (ii) execution time?

Q6. You are the lead designer of a new processor. The processor design and compiler are complete, and now you must decide whether to produce the current design as it stands or spend additional time to improve it. You discuss this problem with your hardware engineering team and arrive at the following options:

a. Leave the design as it stands.

Call this base computer M_{base}. It has a clock rate of 500 MHz, and the following measurements have been made using a simulator (See Table 1)

Table 1		
Instruction Class	**CPI**	**Frequency**
A	2	40%
B	3	25%
C	3	25%
D	5	10%

b. Optimize the hardware

- The hardware team claims that it can improve the processor design to give it a clock rate of 600 MHz. Call this computer M_{opt}. The following measurements (Table 2) were made using a simulator for M_{opt}

I. What is the CPI for each computer?

II. What are the MIPS ratings for M_{base} and M_{opt}?

III. How much faster is M_{opt} than M_{base}?

Table 2		
Instruction Class	**CPI**	**Frequency**
A	2	40%
B	2	25%
C	3	25%
D	4	10%

c. The compiler team has heard about the discussion to enhance the computer discussed. The compiler team proposes to improve the compiler for the computer to further enhance its performance. Call this combination of the improved compiler and the base computer M_{comp}. The instruction improvements from this enhanced compiler have been estimated as follows:

Table 3	
Instruction Class	**Percentage of Instructions Executed Vs. Base Computer**
A	90%
B	90%

C	85%
D	95%

For example, if the base computer executed 500 class A instructions, M_{comp} would execute 0.9 x 500 = 450 class A instructions for the same program.

IV. What is the CPI for M_{comp}?

V. How much faster is M_{comp} than M_{base}?

d. The compiler group points out that it is possible to implement both the hardware improvements and the compiler enhancements.

VI. If both the hardware and compiler improvements are implemented, yielding computer M_{both}, how much faster is M_{both} than M_{base}?

e. You must decide whether to incorporate the hardware enhancements suggested or the compiler enhancements (or both) to the base computer. You estimate that the following time would be required to implement the optimizations:

Table 4	
M_{opt}	6 months
M_{comp}	8 months
M_{both}	8 months

Given that CPU performance improves by approximately 50% per year, or about 3.4% per month. Assuming that the base computer has performance equal to that of its competitors, which optimizations (if any) would you choose to implement?

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